Liu, Shengjie Kris

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Academic Employment

Columbia University in the City of New York
Summer Momentum Fellow, 2025.

The University of Hong Kong
Research Assistant, 2019–2021.

Education

University of Southern California
Doctor of Philosophy, 2021–2026.
Thesis: Deep learning to generate temperature data at high spatiotemporal resolution.

Sun Yat-Sen University
Bachelor of Science, 2015–2019.
Thesis: Deep learning for land use and land cover classification.

Peer-Reviewed Publications, Selected

Google Scholar: D2ZRcjQAAAAJ
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[1] **S. Liu**, L. Zhang, S. Wang, "Comparing temperature data sets for small-area assessment of climate and health: A case study of diarrhea in New York City," *GeoHealth* 10, e2025GH001694, 2026.

[2] **S. Liu**, S. Wang, L. Zhang, "Daily land surface temperature reconstruction in Landsat cross-track areas using deep ensemble learning with uncertainty quantification," *IEEE Trans. Geosci. Remote Sens.* 63, 5011018, 2025.

[3] C.W. So, C.S.J. Pun, **S. Liu**, S.L. Cheung, H.K.K. Hui, K. Blumenthal, C.E. Walker, "Natural experiments from Earth Hour reveal urban night sky being drastically lit up by few decorative buildings," *Sci. Rep.* 15, 21414, 2025.
(Reported in *Forbes*).

[4] **S. Liu**, E. Smith-Greenaway, "Racial and ethnic minorities disproportionately exposed to extreme daily temperature variation in the United States," *PNAS Nexus* 3, pgae176, 2024.
(Featured in *PreventionWeb* [United Nations Office for Disaster Risk Reduction]).

[5] **S. Liu**, L. Zhang, "Deep feature Gaussian processes for single-scene aerosol optical depth reconstruction," *IEEE Geosci. Remote Sens. Lett.* 21, 5505705, 2024.

[6] **S. Liu**, H.C. Ho, "Effects of socioeconomic status and greenspace on respiratory emergency department visits under short-term temperature variations: An age-stratified case time-series study," *Soc. Sci. Med.* 343, 116613, 2024.

[7] C.C.M. Kyba, M. Aubé, S. Bará, A. Bertolo, C.A. Bouroussis, S. Cavazzani, B.R. Espey, F. Falchi, G. Gyuk, A. Jechow, M. Kocifaj, Z. Kolláth, H. Lamphar, N. Levin, **S. Liu**, S.D. Miller, S. Ortolani, C.S.J. Pun, S.J. Ribas, T. Ruhtz, A. Sánchez de Miguel, M. Schneider, R.M. Shrestha, A. Simoneau, C.W. So, T. Storch, K.P. Tong, D. Turnshek, K. Walczak, J. Wang, Z. Wang, J. Zhang, "Multiple angle observations would benefit visible band remote sensing using night lights," *J. Geophys. Res.: Atmos.* 127, e2021JD036382, 2022.
(Reported in *Science*).

[8] **S. Liu**, Q. Shi, L.P. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.* 59, 5085–5102, 2021.
(Clarivate ESI Highly Cited Paper [1%]).

[9] **S. Liu**, Z. Zhou, H. Ding, Y. Zhong, Q. Shi, "Crop mapping using Sentinel full-year dual-polarized SAR data and a CPU-optimized convolutional neural network with two sampling strategies," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 14, 7017–7031, 2021.

[10] **S. Liu**, Q. Shi, "Local climate zone mapping as remote sensing scene classification using deep learning: A case study of metropolitan China," *ISPRS J. Photogramm. Remote Sens.* 164, 229–242, 2020.

Professional Activities & Service

Technical Program Committee: IEEE IGARSS 2025–2026.
Session Chair: IEEE IGARSS 2023–2026.
Scientific Committee: IEEE IGARSS 2023–2026.
Ad Hoc Reviewer for 40+ journals, including *Remote Sensing of Environment*, *IEEE Transactions on Geoscience and Remote Sensing*, *Expert Systems With Applications*, *Pattern Recognition Letters*, *Sustainable Cities and Society*, *Urban Climate*, *Environmental Research*, and *Engineering Applications of Artificial Intelligence*.

Honors & Awards

IEEE Geoscience and Remote Sensing Society Student Paper Award [0.9%, 3/340+], IGARSS 2025.
NSF STC LEAP Summer Momentum Fellowship at Columbia University, 2025.
Climate and Health Science Communication Award Finalist, 2024.
AGU Michael H. Freilich Data Visualization Competition Runner-up Winner, 2024.
USC Dornsife PhD Academy Scholarship, 2022–2025.
IEEE Geoscience and Remote Sensing Society Travel Grant [2018, 2024, 2025].
SYSU EMBA Alumni Association Scholarship, 2018.